

IELTS Academic Reading — Test 01

Time allowed: 60 minutes Total questions: 40

READING PASSAGE 1

You should spend about 20 minutes on Questions 1–13, which are based on Reading Passage 1.

The Science of Sleep

A For most of human history, sleep was regarded as little more than a passive interruption to waking life — an obligatory period of unconsciousness about which science had relatively little to say. That view has been comprehensively overturned. Contemporary sleep research, drawing on neuroimaging, polysomnography, and decades of carefully controlled clinical studies, has established that sleep is among the most biologically active states a human being can occupy. Far from being a time when the brain shuts down, sleep is a period of intense neural activity during which essential processes of memory consolidation, metabolic regulation, cellular repair, and emotional processing are carried out. Scientists now understand that sleep is not a single uniform state but a dynamic cycle of physiologically distinct phases, each serving specific and irreplaceable functions.

B A full sleep cycle lasts approximately ninety minutes and comprises two broad categories: non-rapid eye movement sleep (NREM) and rapid eye movement sleep (REM). NREM sleep itself is divided into three stages of progressively deeper unconsciousness. In Stage 1, the lightest phase, the brain transitions from wakefulness, muscle activity decreases, and individuals may experience hypnic jerks — sudden involuntary muscular contractions. Stage 2 is characterised by the appearance of sleep spindles, brief bursts of high-frequency neural activity that are believed to play a central role in the consolidation of procedural memory and the filtering of sensory information. Stage 3, sometimes called slow-wave or deep sleep, is the phase during which the body carries out its most intensive physical repair work: tissue is regenerated, bone and muscle are built, and growth hormone is secreted by the pituitary gland. REM sleep, which increases in duration across successive cycles throughout the night, is the stage most closely associated with dreaming. During REM, the brain exhibits electrical activity similar to wakefulness, while the voluntary muscles are temporarily paralysed — a phenomenon researchers call REM atonia — preventing the sleeper from physically acting out their dreams.

C The timing and quality of sleep are governed primarily by the body's circadian rhythm, an internal biological clock with a natural cycle of approximately twenty-four hours. This

system is calibrated by environmental cues, most powerfully by light. Specialised photoreceptor cells in the retina detect changes in light exposure and relay this information to the suprachiasmatic nucleus (SCN) in the hypothalamus, the brain region that functions as the master pacemaker of circadian timing. As light levels fall in the evening, the SCN signals the pineal gland to release melatonin, a hormone that promotes feelings of drowsiness and prepares the body for sleep. Conversely, morning light suppresses melatonin secretion and promotes wakefulness. Artificial light from electronic devices — particularly blue-spectrum wavelengths emitted by smartphones and computer screens — can suppress melatonin production in the evening, delaying sleep onset and disrupting the natural sleep-wake cycle.

D The consequences of insufficient sleep are far-reaching and well-documented. In the short term, even a single night of poor sleep measurably impairs attention, working memory, reaction time, and emotional regulation. Sustained sleep deprivation — defined in research literature as consistently obtaining less than six hours per night — is associated with significantly elevated risks of type 2 diabetes, cardiovascular disease, obesity, and depression. A landmark study published in the journal *Science* in 2013 demonstrated that the brain’s glymphatic system — a network of channels surrounding the blood vessels — is dramatically more active during sleep than during wakefulness, flushing out toxic metabolic waste products including beta-amyloid, a protein associated with the development of Alzheimer’s disease. This finding raised the alarming possibility that chronic sleep restriction may accelerate neurodegenerative processes.

E Health authorities including the American Academy of Sleep Medicine and the World Health Organisation recommend that adults aged eighteen to sixty-four obtain between seven and nine hours of sleep per night. Teenagers require eight to ten hours, while children under twelve benefit most from nine to twelve hours per night. Despite these guidelines, surveys conducted across industrialised nations consistently show that a significant proportion of adults report sleeping fewer than seven hours per night. In the United States alone, the Centers for Disease Control and Prevention estimated in 2016 that approximately one in three adults was failing to meet the recommended sleep duration, a pattern described by some researchers as a public health crisis.

F Among the most prevalent modern sleep disorders is insomnia, characterised by persistent difficulty falling asleep, staying asleep, or achieving restorative sleep. Insomnia affects an estimated ten to thirty per cent of adults in developed countries and is closely associated with chronic stress, anxiety, and the demands of shift work. Obstructive sleep apnoea — in which the soft tissues of the throat partially or completely block the airway during sleep, causing repeated interruptions to breathing — is estimated to affect up to one billion people worldwide and is strongly correlated with obesity. Restless legs syndrome, characterised by an irresistible urge to move the legs during rest, affects between five and ten per cent of the adult population in Western nations. The economic cost of sleep disorders, including lost workplace productivity, increased healthcare utilisation, and higher accident rates, has been estimated by the RAND Corporation at over four hundred billion dollars annually in the United States alone.

QUESTIONS 1–13

Questions 1–5

Do the following statements agree with the information given in Reading Passage 1?

Write: - **TRUE** if the statement agrees with the information - **FALSE** if the statement contradicts the information - **NOT GIVEN** if there is no information on this in the passage

1. Stage 2 of NREM sleep is thought to contribute to the consolidation of procedural memory.
2. During REM sleep, the voluntary muscles of the body are temporarily paralysed to prevent sleepers from acting out their dreams.
3. The natural human circadian cycle lasts exactly twenty-four hours under all conditions.
4. Research published in *Science* in 2013 linked the brain's glymphatic system to the removal of waste products including beta-amyloid.
5. People who suffer from obstructive sleep apnoea are more likely than others to experience vivid nightmares during REM sleep.

Questions 6–10

Answer the questions below.

*Choose **NO MORE THAN TWO WORDS** from the passage for each answer.*

6. What is the name of the brain region that serves as the master pacemaker of the body's circadian timing system?
7. What specific type of light emitted by electronic devices is described as being most disruptive to melatonin production?
8. What medical condition is identified in the passage as being strongly correlated with obstructive sleep apnoea?
9. What is the name given to the involuntary muscular contractions that can occur during Stage 1 of NREM sleep?
10. According to the passage, which organisation alongside the World Health Organisation recommends that adults obtain seven to nine hours of sleep per night?

Questions 11–13

Complete the sentences below.

*Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.*

11. A complete sleep cycle lasts approximately minutes from start to finish.
12. During slow-wave sleep, the pituitary gland secretes , which supports physical repair and growth.
13. The RAND Corporation estimated that sleep disorders cost the United States over dollars every year in lost productivity and related expenses.
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READING PASSAGE 2

You should spend about 20 minutes on Questions 14–26, which are based on Reading Passage 2.

Beans of Empire: The Global History of the Coffee Trade

A Few commodities have shaped world history as quietly yet as profoundly as the coffee bean. From its origins in the highlands of northeast Africa to its role as the lubricant of Enlightenment intellectual life and the foundation of a global agricultural industry, coffee has traced a trajectory that mirrors the patterns of exploration, commerce, conquest, and cultural exchange that define the modern world. Today, more than two billion cups of coffee are consumed every day across the globe, making it the world's second most traded commodity after crude oil. Yet the story of how a bitter tropical berry became so central to daily life across every inhabited continent is one that most drinkers have never paused to consider.

B The history of coffee consumption is generally traced to the highlands of Ethiopia, where the *Coffea arabica* plant grows wild in the forested slopes of the Kaffa region. Ethiopian legend credits a goat herder named Kaldi with the accidental discovery of coffee's stimulating properties sometime in the ninth century, after noticing that his animals became unusually energetic following their consumption of bright red berries from an unfamiliar shrub. While

the Kaldi legend is almost certainly apocryphal, there is credible historical and archaeological evidence that coffee plants were being cultivated and consumed in Ethiopia by at least the fifteenth century. Early Ethiopian consumption was not in the form of the brewed liquid familiar today; rather, the berries and leaves were often eaten raw, mixed with animal fat into portable energy balls, or brewed into a fermented alcoholic drink known as qishr.

C It was the trade routes across the Red Sea that carried the coffee plant from Ethiopia to the Arabian Peninsula, where the city of Mocha on the coast of Yemen became the world's first major centre of coffee cultivation and commerce. Sufi monks were among the earliest enthusiastic adopters of the beverage in its brewed form, consuming coffee during night-time religious devotions to maintain wakefulness and enhance their powers of concentration. By the mid-fifteenth century, coffeehouses — known as qahveh khaneh — had appeared in Mecca and Medina, and they quickly became central institutions of social and intellectual life, drawing merchants, scholars, musicians, and storytellers. The Ottoman governors of the region became so alarmed by the political discussions that took place in these establishments that in 1511 the Governor of Mecca, Khair Beg, attempted to prohibit both coffee and coffeehouses on the grounds that they encouraged seditious conversation. His ban was overturned within months by the Sultan in Cairo, who reportedly enjoyed the beverage himself.

D Coffee reached Constantinople, the capital of the Ottoman Empire, in 1555, when two Syrian merchants named Hakm and Shams opened the first of what would become hundreds of coffeehouses in the city. Ottoman coffeehouses became celebrated centres of social life, where men gathered to play chess, listen to music, exchange news, and debate politics and philosophy — activities that earned them the informal title of “schools of the wise.” Coffee spread rapidly through the Ottoman territories into Persia, Egypt, and North Africa across the sixteenth century, carried by trade caravans and returning pilgrims from Mecca. By the early seventeenth century, Venetian traders had introduced coffee to Europe, where it encountered initially fierce resistance from the Catholic Church. Pope Clement VIII, urged by advisers to condemn coffee as a Muslim beverage, is said to have tasted it and found it so delicious that he gave it his papal blessing instead — though this story, like many in coffee history, may be more folklore than fact.

E Coffeehouses arrived in Oxford in 1650, in London in 1652, and spread rapidly across Britain, France, and the Netherlands over the following decades. London alone had over three hundred coffeehouses by 1675. Unlike taverns, which served alcohol and were associated with disorder and licentiousness, coffeehouses were sober institutions that charged a penny for admission and provided newspapers and an environment conducive to business discussion. Lloyd's of London — the famous insurance market — originated in Edward Lloyd's coffeehouse near the Thames, where ship captains, merchants, and underwriters gathered to exchange maritime intelligence and conduct transactions. The stock market equivalent, Jonathan's Coffee House, became the London Stock Exchange. Coffeehouses thus played a direct role in the development of modern financial institutions.

F The insatiable European demand for coffee drove a dramatic expansion of colonial plantation agriculture. The Dutch, who had obtained coffee seedlings from Yemen and established

the world's first colonial coffee plantations in Java in 1696, subsequently transplanted them to Suriname in South America. The French brought coffee cultivation to Martinique and Guadeloupe in the Caribbean, and from there to Saint-Domingue — now Haiti — which by the mid-eighteenth century was supplying over half the world's coffee. Brazil, which today accounts for approximately thirty-five per cent of global coffee production, received its first coffee plants from French Guiana in 1727 through a scheme of considerable diplomatic audacity involving a Portuguese military officer named Francisco de Melo Palheta, who allegedly charmed the wife of the Governor of French Guiana into providing him with fertile coffee seedlings as a parting gift. The plantation systems that produced this coffee were built overwhelmingly on the labour of enslaved Africans, a fact that connects the pleasures of the morning cup to one of the most brutal chapters in the history of human exploitation.

G In the contemporary global economy, coffee supports the livelihoods of an estimated twenty-five million smallholder farmers across the tropical belt of Africa, Asia, and Latin America. The industry has faced sustained criticism for the low prices typically paid to producers relative to the retail prices commanded by branded coffee in consuming countries. The fair trade and direct trade movements that emerged in the 1980s and 1990s sought to address this imbalance by guaranteeing minimum prices to growers and establishing more equitable supply chains. Climate change now presents a fundamental threat to global coffee production: rising temperatures, shifting rainfall patterns, and the expansion of fungal diseases such as coffee leaf rust are projected to reduce the area of land suitable for Arabica cultivation by up to fifty per cent by 2050, raising questions about the long-term future of one of the world's most beloved beverages.

QUESTIONS 14–26

Questions 14–18

Reading Passage 2 has seven paragraphs, A–G.

Choose the correct heading for paragraphs B–F from the list of headings below.

Write the correct Roman numeral, i–x, next to Questions 14–18.

List of Headings

i	Coffee's arrival in Europe and the birth of the coffeehouse culture
ii	Trade, conquest, and colonial plantation labour

iii	The legendary African origins of the coffee plant
iv	Religious authorities attempt to suppress the coffee trade
v	The Arabian Peninsula as the birthplace of commercial coffee culture
vi	Coffee's expansion through the Ottoman Empire
vii	How coffeehouses shaped European commerce and finance
viii	The modern fair trade movement and climate threats
ix	Scientific studies into coffee's health benefits
x	Coffee's roots in Ethiopian agriculture and early consumption

14. Paragraph B

15. Paragraph C

16. Paragraph D

17. Paragraph E

18. Paragraph F

Questions 19–22

Choose the correct letter, A, B, C or D.

19. According to the passage, in what form was coffee most commonly consumed in early Ethiopia?

- **A** Brewed as a hot liquid similar to modern coffee
- **B** Eaten raw or mixed with fat, or fermented into an alcoholic drink
- **C** Ground into a powder and dissolved in cold water
- **D** Dried and smoked as part of religious ceremonies

20. Why did the Governor of Mecca attempt to ban coffeehouses in 1511?

- **A** He believed coffee was harmful to physical health.

- **B** He objected to the religious nature of conversations held there.
- **C** He was concerned that political discussions in coffeehouses were dangerous.
- **D** He wanted to protect the economic interests of local tea merchants.

21. What was the significance of Lloyd's of London's origins in a coffeehouse?

- **A** It demonstrated that coffeehouses functioned as informal universities.
- **B** It showed that coffeehouses played a direct role in the development of modern financial institutions.
- **C** It illustrated how coffeehouses competed directly with taverns for customers.
- **D** It proved that coffee consumption increased productivity in commercial settings.

22. According to paragraph F, how did Brazil first obtain its coffee plants?

- **A** They were purchased officially from Dutch colonial authorities.
- **B** They were obtained through an arrangement involving coffee seedlings provided by a French Guianan official's wife.
- **C** They arrived via slave traders who brought them from Africa.
- **D** They were developed through independent cross-breeding experiments by Brazilian scientists.

Questions 23–26

Complete the sentences below.

*Choose **NO MORE THAN TWO WORDS** from the passage for each answer.*

23. In Yemen, Sufi monks were among the first to drink coffee regularly, using it to maintain during night-time religious devotions.

24. When coffeehouses spread through Europe, they were regarded as more respectable than taverns because they were sober institutions associated with rather than with disorder.

25. The Dutch established the world's first colonial coffee plantations in in 1696 after obtaining coffee seedlings from Yemen.

26. Environmental threats to coffee production include rising temperatures and the spread of fungal diseases such as coffee

READING PASSAGE 3

You should spend about 20 minutes on Questions 27–40, which are based on Reading Passage 3.

Thinking Fast and Slow: Cognitive Psychology and the Science of Decision-Making

A The capacity to make decisions is so fundamental to human cognition that it is easy to assume the process is largely rational, consistent, and well-calibrated. For much of the twentieth century, economists and policy-makers operated on precisely this assumption: that individuals, given relevant information and sufficient time, would weigh options in a systematic fashion and select the outcome that maximised their welfare. This model, variously known as expected utility theory or rational choice theory, underpinned generations of economic modelling, legal reasoning, and public policy. The uncomfortable truth that has emerged from decades of research in cognitive psychology is that human decision-making departs systematically and predictably from the rational ideal — and that understanding these departures has profound implications for economics, medicine, law, and governance.

B The most influential framework for understanding the architecture of human thought was developed by Israeli-American psychologist Daniel Kahneman, who synthesised decades of research in his 2011 book *Thinking, Fast and Slow*. Kahneman proposed that the mind operates through two broadly distinct modes, which he labelled System 1 and System 2. System 1 is fast, automatic, and largely unconscious: it generates intuitions, impressions, and emotional reactions with minimal deliberate effort, drawing on pattern recognition honed by experience. System 2 is slow, deliberate, and effortful: it is engaged when we perform complex calculations, evaluate logical arguments, or carefully weigh competing alternatives. The critical insight is that System 1 is the dominant mode of operation in everyday life; System 2 is comparatively lazy and tends to endorse the impressions and intuitions generated by System 1 rather than subjecting them to independent critical scrutiny.

C A central finding of cognitive psychology is that human beings rely on mental shortcuts — known as heuristics — when making judgements under uncertainty. Heuristics are efficient rules of thumb that generally produce adequate results and conserve cognitive resources. The availability heuristic, identified by Kahneman and his long-term collaborator Amos Tversky, causes people to judge the frequency or probability of events based on how easily relevant examples come to mind. Events that are vivid, recent, or emotionally charged are more easily recalled and are therefore systematically overestimated in terms of their likelihood. The anchoring heuristic leads individuals to rely disproportionately on the first piece of numerical information they encounter — the “anchor” — when making subsequent estimates. Research has shown that even patently arbitrary anchors — such as a number generated by spinning a roulette wheel — can exert a measurable influence on people’s numerical judgements.

D Heuristics are closely related to cognitive biases — systematic and predictable errors in reasoning that arise from the use of mental shortcuts in contexts where they are ill-suited. The confirmation bias describes the tendency to seek out, interpret, and remember information in ways that confirm one’s pre-existing beliefs while disregarding or downweighting information that challenges them. The optimism bias leads individuals to systematically underestimate the probability of adverse outcomes — illness, accident, financial loss — and overestimate the likelihood of positive ones. The sunk cost fallacy causes people to continue investing time, money, or effort in a failing project because of what has already been expended on it, rather than evaluating current prospects on their own merits. These biases are not random errors; they are consistent, reproducible, and found across diverse cultural contexts.

E A foundational concept in the psychology of decision-making is bounded rationality, a term introduced by the American polymath Herbert Simon in the 1950s. Simon argued that the classical economic model of rational choice was psychologically unrealistic: it assumed that decision-makers had access to complete information, unlimited computational capacity, and ample time to evaluate all possible options. In practice, human beings operate under severe constraints on all three fronts. In response to these constraints, Simon argued, people do not optimise — they satisfice, selecting the first option that meets a threshold of acceptability rather than exhausting all possibilities in search of the mathematically optimal solution. The concept of bounded rationality was a foundational contribution to what would later become the field of behavioural economics.

F Perhaps the most celebrated contribution of Kahneman and Tversky to the psychology of decision-making was prospect theory, first presented in a landmark 1979 paper in the journal *Econometrica*. Prospect theory describes how individuals actually evaluate potential gains and losses, and departs sharply from classical utility theory in several key respects. Most significantly, it demonstrates that losses loom larger than equivalent gains in psychological terms: the pain of losing one hundred dollars is felt approximately twice as intensely as the pleasure of gaining the same amount, a phenomenon known as loss aversion. Furthermore, the psychological impact of additional gains or losses diminishes as their absolute magnitude increases — a property referred to as the diminishing sensitivity of the value function. Prospect theory has since been replicated across hundreds of studies in dozens of countries, and its authors received the Nobel Prize in Economic Sciences in 2002.

G The implications of cognitive psychology for economics have been transformative. The field of behavioural economics, which integrates psychological findings with economic theory, has demonstrated that many standard assumptions of classical economics — including the idea that individuals make consistent, preference-satisfying choices — are empirically false. Behavioural economists have shown that choices are sensitive to the way options are framed: identical outcomes described in terms of gains and in terms of losses produce systematically different choices. They have also demonstrated that individuals value goods they already possess more highly than equivalent goods they do not own — the endowment effect — and that social comparisons profoundly shape what people regard as fair, acceptable, or desirable. These

insights have been used to redesign savings schemes, healthcare programmes, and consumer protection regulations.

H The policy application of insights from cognitive psychology is most closely associated with the concept of nudging, which was popularised by economists Richard Thaler and Cass Sunstein in their 2008 book *Nudge*. A nudge is any feature of a decision environment that predictably alters behaviour without forbidding options or significantly changing financial incentives. Classic examples include placing healthier foods at eye level in workplace canteens, automatically enrolling employees in pension savings schemes with an opt-out option rather than an opt-in, and configuring organ donation systems as opt-out rather than opt-in. Research across multiple countries has demonstrated that such architecture interventions — which exploit the human tendency towards inertia — can produce substantial changes in behaviour at negligible cost. Critics, however, have questioned whether nudging is paternalistic, whether it manipulates people without their knowledge or consent, and whether it addresses only the symptoms of irrational decision-making while leaving its underlying causes unaddressed.

QUESTIONS 27–40

Questions 27–31

Reading Passage 3 has eight paragraphs, A–H.

Which paragraph contains the following information?

Write the correct letter, A–H, next to Questions 27–31.

NB: You may use any letter more than once.

- 27.** An explanation of why people tend to overestimate the probability of dramatic or recent events.
 - 28.** A reference to research showing that losses are felt more acutely than equivalent gains.
 - 29.** An account of the argument that people choose “good enough” options rather than the best possible option.
 - 30.** An example of a policy intervention that uses the design of a choice environment to change behaviour without removing options.
 - 31.** A description of research showing that the order in which information is presented influences people’s numerical estimates.
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Questions 32–36

Do the following statements agree with the information given in Reading Passage 3?

Write: - **TRUE** if the statement agrees with the information - **FALSE** if the statement contradicts the information - **NOT GIVEN** if there is no information on this in the passage

32. According to Kahneman's framework, System 2 thinking typically overrides the conclusions produced by System 1 in most everyday situations.

33. Kahneman and Tversky published their prospect theory paper in the journal *Econometrica* in 1979.

34. The sunk cost fallacy involves making decisions based on the likelihood of future success rather than past expenditure.

35. Prospect theory has been tested in a limited number of countries due to difficulties in cross-cultural research design.

36. Richard Thaler and Cass Sunstein co-authored the book *Nudge*, which was published in 2008.

Questions 37–40

Answer the questions below.

*Choose **NO MORE THAN THREE WORDS** from the passage for each answer.*

37. What term did Daniel Kahneman use to describe the rapid, automatic, and largely unconscious mode of human thought?

38. According to the passage, what economic phenomenon describes people placing a higher value on goods they already own than on equivalent goods they do not own?

39. What was the name of the book in which Daniel Kahneman synthesised his decades of research for a general audience?

40. What term did Herbert Simon use to describe the process of selecting the first acceptable option rather than searching exhaustively for the optimal solution?

END OF TEST 01 — READING